

Arithmetic Integers

Review Exercise Answers

Level-2

S1. We have:

$$S = n^5 - n = n(n^4 - 1) = n(n^2 - 1)(n^2 + 1)$$

$$\Rightarrow S = (n-1)(n)(n+1)(n^2 + 1)$$

The product of any three consecutive integers is always a multiple of 6 (how?). This means that S above is a multiple of 6, since $n-1$, n and $n+1$ are consecutive integers. Now, we show that S is also a multiple of 5. If any of the terms $n-1$ or n or $n+1$ is a multiple of 5, then we are done. But suppose that none of these is a multiple of 5. This means that n must be of the form $5k+2$ or $5k+3$. Can you see how? This step is important, so make sure you understand it. The next step is to show that if n is of the form $5k+2$ or $5k+3$, then n^2+1 will be a multiple of 5. This is easy, and once again, left to you as an exercise. In all cases, S is a multiple of 5. Since it is also a multiple of 6, S is a multiple of $5 \times 6 = 30$.

S2. The given statement is false. We have:

$$S = n^4 - 1 = (n^2 - 1)(n^2 + 1)$$

$$= (n-1)(n+1)(n^2 + 1)$$

Since n is odd, we let $n = 2m+1$. Thus,

$$S = (2m)(2m+2)(4m^2 + 4m + 2)$$

$$= 4m(m+1)(4m(m+1) + 2)$$

The term $4m(m+1)$ is a multiple of 8. However, the expression in the second bracket is $4m(m+1) + 2$, which will be of the form $4k+2$. Thus, it is never a multiple of 4. This means that S is not necessarily a multiple of $8 \times 4 = 32$.

S3. Note that $n^5 + 4n = n(n^4 + 4)$. If n itself is a multiple of 5, then we are done. Otherwise, we have to show that $n^4 + 4$ must be a multiple of 5. Now, if n is not a multiple of 5, we take $n = 5k+r$, where r is in the set $\{1, 2, 3, 4\}$. For each of these four values, we can show that $n^4 + 4$ will be a multiple of 5. For example, if we take $r=1$, we have:

$$n^4 + 4 = (5k+1)^4 + 4$$

$$= 5m + 1 + 4 = 5m + 5 = 5m'$$

We have made use of the fact that expanding $(5k+1)^4$ will yield a number of the form $5m+1$ (you can verify this). Similarly, if r is 2, we have:

$$\begin{aligned} n^4 + 4 &= (5k+2)^4 + 4 \\ &= 5m + 16 + 4 = 5m + 20 = 5m' \end{aligned}$$

Thus, in each value for r , $n^4 + 4$ will turn out to be a multiple of 5. You are urged to complete the proof yourself.

- S4.** We will provide the solution to one part and leave the rest to you as an exercise. Note that we can write the following:

$$\begin{aligned} a &= k_1n + r_1 \\ b &= k_2n + r_2 \\ \Rightarrow a + b &= (k_1 + k_2)n + (r_1 + r_2) \end{aligned}$$

Clearly, the LHS $a + b$ will leave the same remainder upon division by n as the second term on the RHS, which is $r_1 + r_2$. The other parts can be proved similarly.

- S5.** Self-exercise.

- S6.** We will show that the GCD of the numerator and the denominator is 1. We have:

$$(12n+1)(5) + (30n+2)(-2) = 1$$

Thus, the fraction is in its simplest form.

- S7.** (a) First, we note that any prime number greater than 3 will either be of the form $6k-1$ or $6k+1$. Your first task is to prove this fact.

Thus, we can assume that $m = 6k \pm 1$ for some integer k . Now,

$$\begin{aligned} m^2 - 1 &= (6k \pm 1)^2 - 1 \\ &= (36k^2 \pm 12k + 1 - 1) \\ &= 12k(3k \pm 1) \end{aligned}$$

The term $k(3k \pm 1)$ will always be even, which means that $m^2 - 1$ must be a multiple of 24.

(b) By the result of part-(a), both $m^2 - 1$ and $n^2 - 1$ are multiples of 24, and thus so is their difference, which is $m^2 - n^2$.

- S8.** (a) The first few powers of 2 are:

2, 4, 8, 16, 32, 64, 128, 256, ...

Clearly, the last digit of the powers has the following cycle:

$\underbrace{2, 4, 8, 6}_{\text{cycle}}, 2, 4, 8, 6, \dots$

Thus, 2^{100} will have a last digit of 6.

- (b) First, understand that the last digit of 77^{77} will be the same as the last digit of 7^{77} . Now, we look at the first few powers of 7:

7, 49, 343, 2401, 16807, ...

The last digit has the following cycle: 7, 9, 3, 1. Thus, 7^{77} will have its last digit has the first number of this cycle, or 7.

- S9.** If neither a nor b is divisible by 3, then both of them are of the form $3k \pm 1$, so that their squares will be of the form $3p + 1$. Thus, their sum will be of the form $3q + 2$. However, no perfect square can be of the form $3q + 2$ (why?).

- S10.** First, show that if a number p leaves a remainder of r when divided by 6, then p^3 will also leave the same remainder when divided by 6. For example, suppose that p leaves a remainder of 2 when divided by 6. We can write $p = 6k + 2$, and so:

$$p = 6k + 2$$

$$p^3 = (6k + 2)^3$$

$$= 6k^3 + 8 = 6k^3 + 6 + 2$$

$$= 6l + 2$$

Similarly, show for the other cases that p and p^3 leave the same remainder when divided by 6.

Now, let r_1, r_2, r_3 be the remainders when x, y, z are divided by 6 respectively. We can write:

$$x = 6k_1 + r_1$$

$$y = 6k_2 + r_2$$

$$z = 6k_3 + r_3$$

$$\Rightarrow x + y + z = 6(k_1 + k_2 + k_3)$$

$$+ (r_1 + r_2 + r_3)$$

$$= 6k + (r_1 + r_2 + r_3)$$

If $x + y + z$ is divisible by 6, this means that $r_1 + r_2 + r_3$ is also divisible by 6.

Now, x^3, y^3, z^3 will also leave remainders r_1, r_2, r_3 when divided by 6. Thus, their sum will leave the same remainder when divided by 6 as $r_1 + r_2 + r_3$. But since we have shown that $r_1 + r_2 + r_3$ is divisible by 6, we can conclude that $x^3 + y^3 + z^3$ is divisible by 6.